

Expo-GS: Exposure-Aware Signed Distance Function in Gaussian Splatting for High Dynamic Range

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Abstract

High dynamic range novel view synthesis (HDR-NVS) remains challenged by geometric artifacts and radiometric distortions under multi-exposure conditions, primarily due to existing methods ignoring exposure and over-relying on color cues. Inspired by the integrated processing of color and structure of the human visual system (HVS), we propose Expo-GS, a novel framework that decomposes HDR-NVS into three interpretable components, namely, Irradiance Field Training, Geometry Field Training, and Interactive Joint Training. Central to Expo-GS is the exposure-aware signed distance function (Expo-SDF), which dynamically reweights geometric supervision via localized exposure reliability estimation, suppressing noisy gradients from unstable regions while enhancing structure learning in well-exposed areas. Building on this, we design an interactive optimization strategy that synchronizes Gaussian primitive growth and pruning with evolving Expo-SDF cues, enabling exposure-aware density control and eliminating hallucinated structures near exposure transitions. Experiments show that Expo-GS significantly outperforms prior methods on both synthetic and real-world datasets. It achieves a peak PSNR of 39.06 dB under HDR settings and up to 41.38 dB in the LDR-OE configuration, excelling in preserving high-frequency textures and maintaining structural consistency.

1. Introduction

Novel view synthesis (NVS) aims to generate photorealistic renderings from sparse input views (Dalal et al., 2024), enabling continuous viewpoint interpolation for applications

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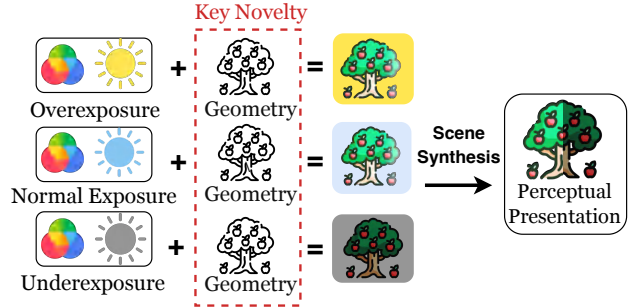


Figure 1. An Overview of the Motivation. Our method draws inspiration from the human visual system (HVS), which decomposes visual input into complementary cues such as color and geometry. Following this principle, our scene-synthesis framework aggregates appearance information across multiple exposures and fuses it with geometric information, enabling the model to form a more comprehensive representation of the scene.

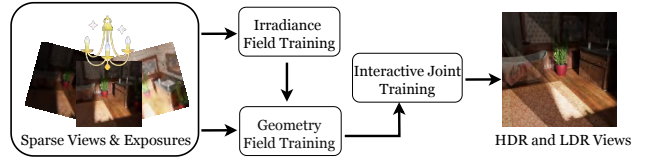


Figure 2. An Overview of the Expo-GS Pipeline. Inspired by the HVS, our method devises a unified pipeline comprising irradiance field, geometry field, and interactive joint training to jointly model radiance and geometry across multi-exposure conditions.

in virtual reality (Xu et al., 2023), autonomous systems (Hess et al., 2025), and creation of three-dimensional content (Tang et al., 2023). However, conventional 8-bit low dynamic range (LDR) imaging struggles to capture the full radiometric complexity of real-world scenes, especially under extreme exposure, leading to degraded perceptual and geometric fidelity. In contrast, high dynamic range (HDR) imaging employs substantially higher per-channel bit depth to more faithfully capture the physical radiance, thereby mitigating the artifacts introduced by conventional nonlinear compression (Chen et al., 2025).

Existing high dynamic range novel view synthesis (HDR-NVS) approaches can be broadly divided into two categories: NeRF-based methods (Mildenhall et al., 2021;

(Martin-Brualla et al., 2021; Huang et al., 2022)) and 3D Gaussian Splatting (3D-GS) techniques (Kerbl et al., 2023; Cai et al., 2024)). NeRF-based models encode scene radiance as continuous volumetric fields, enabling the synthesis of intricate lighting phenomena. Despite their expressiveness, these methods often fail to recover fine structures in shadowed or underexposed regions and suffer from heavy computational demands due to dense ray sampling. In contrast, 3D-GS offers higher rendering efficiency and has recently been adapted for NVS (Fei et al., 2024). However, its design remains fundamentally tailored to LDR scenarios. Without explicit mechanisms to account for extreme exposure variations tend to generate hallucinated geometry and ghosting artifacts under challenging illumination conditions.

Current HDR-NVS approaches predominantly emphasize irradiance field modeling (Cai et al., 2024), often relying on color regression as the primary strategy to approximate scene appearance under varying exposure conditions (Huang et al., 2022). However, this single-modal learning paradigm poses two critical limitations: (i) Optimization driven solely by color cues fails to capture geometric variations induced by exposure disparities; and (ii) Regions with sharp luminance transitions, such as boundaries between illuminated and shadowed areas, are particularly vulnerable to radiance field distortions due to insufficient structural priors. Inspired by the human visual system (HVS), which functionally decouples color and geometry but integrates them for coherent perception, we introduce a biologically inspired framework (Figure 1) that jointly models irradiance and geometry.

In this paper, we present a novel framework, *Exposure-Aware Signed Distance Function in Gaussian Splatting* (*Expo-GS*), which draws inspiration from the functional organization of the HVS. To address the limitations of existing HDR-NVS approaches, we introduce a disentangled-then-joint training paradigm that adaptively modulates radiometric and geometric supervision based on localized exposure conditions. Expo-GS seamlessly integrates color sensitivity and structural awareness into the GS pipeline, enabling accurate HDR scene reconstruction and robust performance in NVS. Unlike conventional SDF-based methods, our exposure-aware variant dynamically reweights geometric constraints according to the exposure reliability of observed regions. This adaptive mechanism mitigates overfitting in saturated or underexposed areas by down-weighting unreliable radiometric cues, while reinforcing supervision in well-exposed, structurally consistent regions, thus promoting stable and precise geometry estimation under extreme lighting conditions. To model radiance distribution, we adopt a soft forward cumulative rendering strategy during irradiance field training, facilitating efficient Gaussian projection and consistent scene visualization. Moreover, we introduce a joint optimization module to simultaneously refine color fidelity and geometric coherence, fostering a

dynamic balance between radiometric precision and structural integrity. Together, these innovations enable Expo-GS to achieve high-fidelity reconstruction and strong generalization in complex HDR environments, and an overview of the pipeline is shown in Figure 2.

Our main contributions are summarized as follows:

- We propose a novel framework **Expo-GS** that, for the first time, conceptually decomposes HDR-NVS into three disentangled components: color, geometry, and exposure. This factorization improves HDR scene interpretability and supports precise, modular optimization.
- We introduce an exposure-aware signed distance function (**Expo-SDF**) that adaptively integrates geometric supervision across multi-exposure inputs. This mechanism enables reliable structural learning and significantly enhances geometric fidelity and consistency.
- We design a joint optimization strategy that couples Expo-SDF-guided geometry with refined irradiance reconstruction, enabling optimization and producing sharper, more coherent results through a balanced integration of radiance fidelity and geometric stability.

2. Method

2.1. Irradiance Field for Basic Color Regression

We adopt anisotropic Gaussians as the core representational primitives for HDR scene modeling, serving as fundamental units that enable efficient radiance field and geometric field representation:

$$\mathcal{G} = \{g_i = (\boldsymbol{\mu}_i, \boldsymbol{\Sigma}_i, \boldsymbol{\alpha}_i, k_i, \mathcal{M}_\theta)\} \quad (1)$$

where $\boldsymbol{\mu}_i$ is the Gaussian center, $\boldsymbol{\Sigma}_i$ the anisotropic covariance, $\boldsymbol{\alpha}_i$ the radiance coefficients, k_i the scale factor, and $\mathcal{M}_\theta$ the shared SH parameters for radiance modeling. This formulation enables sub-pixel precision and high-frequency encoding under HDR conditions, while remaining compatible with LDR representations. To capture view-dependent HDR radiance, each Gaussian is assigned compact SH coefficients, yielding the observed radiance $\mathbf{c}_i^{\text{HDR}} \in \mathbb{R}^3$ from direction $\mathbf{v} \in \mathbb{S}^2$:

$$\mathbf{c}_i^{\text{HDR}}(\mathbf{v}) = \exp(\mathbf{B}(\mathbf{v})^\top \mathbf{k}_i) \quad (2)$$

Here, $\mathbf{B}(\mathbf{v}) \in \mathbb{R}^{(L+1)^2}$ denotes the real spherical harmonics (SH) basis evaluated at direction $\mathbf{v}$, and $\mathbf{k}_i \in \mathbb{R}^{(L+1)^2 \times 3}$ are the SH coefficients for the i -th point. To convert HDR radiance into LDR images, we employ a learnable tone mapping network that models the non-linear camera response

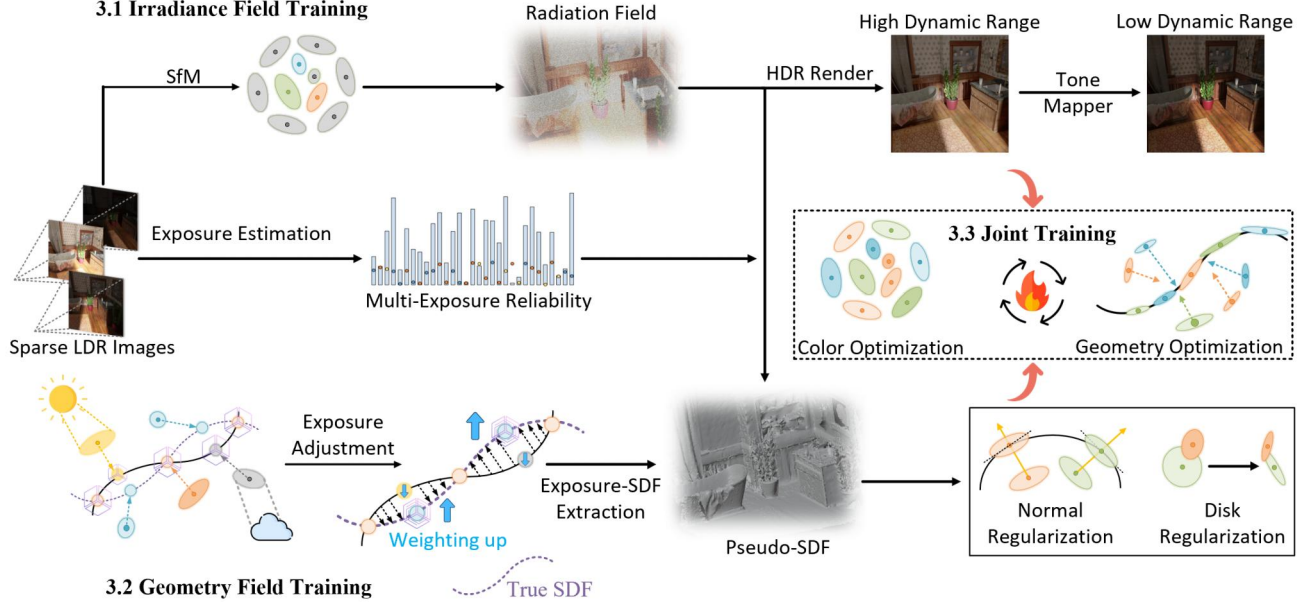


Figure 3. The Expo-GS pipeline. Taking sparse multi-exposure LDR images as input, our framework decomposes HDR-NVS into three interpretable stages: irradiance field training for HDR radiance regression with tone mapping, geometry field training for exposure-aware SDF extraction via reliability-based weighting, and interactive joint training that mutually refines color and geometry through density control and regularization, ultimately synthesizing both HDR and LDR novel views with high fidelity and geometric consistency.

under varying exposures. Instead of using a fixed function, we define tone mapping as a learnable transformation $\mathcal{M}_\theta : \mathbb{R}^3 \rightarrow \mathbb{R}^3$ operating in the log domain:

$$\mathbf{c}_i^{\text{LDR}}(\Delta t) = \mathcal{M}_\theta(\log \mathbf{c}_i^{\text{HDR}} + \log \Delta t) \quad (3)$$

Here, Δt denotes the exposure time and the logarithmic formulation stabilizes the dynamic range to facilitate learning. The network $\mathcal{M}_\theta$ is shared across all Gaussians and channels. We adopt a parallel, fully differentiable rasterization pipeline to project 3D Gaussians onto the 2D image plane. This simulates volumetric accumulation of semitransparent Gaussians. For a pixel p at exposure time Δt , its color is computed using front-to-back soft blending of all overlapping Gaussians:

$$I(p | \Delta t) = \sum_{j \in \mathcal{N}_p} \mathbf{c}_j(\Delta t) \cdot \sigma_j \cdot \prod_{k=1}^{j-1} (1 - \sigma_k) \quad (4)$$

In this expression, $\mathcal{N}_p$ denotes the depth-sorted set of Gaussians visible at pixel p . The term $\mathbf{c}_j(\Delta t) \in \mathbb{R}^3$ is the RGB radiance of the j -th Gaussian, modulated by exposure time via the learned tone mapping function. The visibility weight $\sigma_j \in [0, 1]$ encodes both opacity and projected spatial density. To model soft visibility and volumetric contribution, each Gaussian’s support in screen space is represented by a continuous kernel. The visibility weight σ_j at pixel p is

defined as:

$$\sigma_j = \alpha_j \cdot \exp \left(-\frac{1}{2} (p - \boldsymbol{\mu}_j)^\top \boldsymbol{\Sigma}_j^{-1} (p - \boldsymbol{\mu}_j) \right) \quad (5)$$

where α_j is a learnable opacity scalar determining the base contribution of the Gaussian, and the exponential term models a 2D anisotropic Gaussian kernel centered at $\boldsymbol{\mu}_j$, shaped by $\boldsymbol{\Sigma}_j$. This formulation supports smooth, differentiable visibility estimation and compactly simulates soft occlusion during rasterization. We adopt a unified photometric loss that supports both HDR and tone-mapped LDR supervision, enabling flexible optimization across radiometric and perceptual domains:

$$\mathcal{L}_{\text{render}} = \sum_{j=1}^B [\mathcal{L}_1(\mathcal{R}_j, \mathcal{R}_j^{\text{gt}}) + \lambda \cdot \mathcal{L}_{\text{D-SSIM}}(\mathcal{R}_j, \mathcal{R}_j^{\text{gt}})] \quad (6)$$

where $\mathcal{R}_j$ is the rendered output at the j -th viewpoint, and $\mathcal{R}_j^{\text{gt}}$ is the corresponding ground truth, either in HDR or tone-mapped LDR format depending on the training stage. This formulation is robust to exposure variations and promotes alignment in both pixel-wise and perceptual spaces.

2.2. Geometry Field for Exposure-Aware Structural Modeling

To better model continuous geometric structures in HDR scenes, we propose an exposure-aware signed distance field

(Expo-SDF) informed by Gaussian visibility. Prior work has shown that explicit geometric supervision improves irradiance field optimization. Building on this, we introduce an Expo-SDF-based density formulation that attenuates contributions from overexposed or underexposed regions, while emphasizing well-exposed areas to enhance geometric fidelity. To integrate multi-exposure reliability into the geometry field, we first define an exposure estimation function:

$$\mathcal{E}_i(q) = e_i \cdot \max_{c \in \{R, G, B\}} I_{i,c}(q), \quad (7)$$

where q denotes a pixel coordinate in the LDR image I_i , e_i is the exposure factor (exposure time) for image i , and $I_{i,c}(q)$ is the intensity of channel c . This formulation provides a simple yet effective proxy for assessing the radiometric reliability of individual pixels. Additional design analysis is provided in appendix D. We then project each Gaussian center μ_j into image i to retrieve its exposure estimate as $\mathcal{E}(\mu_j) = \mathcal{E}_i(\pi_i(\mu_j))$, where $\pi_i(\cdot)$ denotes the projection function. Using these estimates, we define an exposure-normalized density function that down-weights unreliable contributions:

$$\dot{d}(p) = \sum_j \frac{\alpha_j}{\mathcal{E}(\mu_j) + \epsilon} \cdot \exp\left(-\frac{1}{2}(p - \mu_j)^\top \Sigma_j^{-1}(p - \mu_j)\right), \quad (8)$$

where $\dot{d}(p)$ denotes the exposure-normalized Gaussian density at 3D location p , α_j is the learnable opacity scalar of the j -th Gaussian, $\mu_j \in \mathbb{R}^3$ is its 3D center, $\Sigma_j \in \mathbb{R}^{3 \times 3}$ is the anisotropic covariance matrix encoding its spatial extent and orientation, $\mathcal{E}(\mu_j)$ is the projected exposure value as defined in Eq. (7), and ϵ is a small constant to avoid division by zero. This formulation effectively suppresses the influence of Gaussians located in radiometrically unreliable regions, while preserving the contributions from well-exposed observations. The derivation and proof are provided in appendix B. To estimate surface proximity in regions lacking reliable exposure or dense geometry supervision, we construct a pseudo-Expo-SDF based on the aggregated Gaussian density:

$$f_{\text{HDR}}(p) = \pm s_{g^*} \cdot \sqrt{-2 \log \dot{d}(p)}, \quad (9)$$

where g^* denotes the index of a locally dominant Gaussian around p under the standard front-to-back compositing, and s_{g^*} is a local scale factor derived from its spatial extent. Both s_{g^*} and the associated normal direction are only used to provide a local scale and orientation for interpreting the smooth potential $-\log \dot{d}(p)$ as a distance-like quantity; they do not modify the underlying log-density field. This formulation provides a smooth and differentiable approximation of the underlying geometry, even in regions with unreliable exposure, and serves as a practical surrogate for mesh initialization, SDF alignment, and supervision pruning. To

ensure geometric consistency between the estimated pseudo-Expo-SDF and the ground-truth surface, we introduce an exposure-aware supervision loss:

$$\mathcal{L}_{\text{SDF}}^{\text{HDR}} = \frac{1}{|P|} \sum_{p \in P} \left| \hat{f}(p) - f_{\text{HDR}}(p) \right|, \quad (10)$$

where $\hat{f}(p)$ denotes the reference signed distance value at point p , obtained via mesh-based rendering or camera-projected depth, and $f_{\text{HDR}}(p)$ is the pseudo-Expo-SDF derived from exposure-aware Gaussian density. This loss enables soft geometric supervision in radiometrically challenging areas, leveraging structural priors without requiring complete or noise-free ground-truth annotations. To further guide geometry learning in radiometrically ambiguous regions, we regularize the pseudo-Expo-SDF field with two complementary constraints: normal alignment and spatial flatness. Specifically, we encourage the gradient of the Expo-SDF field at each point $p \in P$ to align with the dominant surface normal $\mathbf{n}_{g^*}$, promoting local planar consistency:

$$\mathcal{L}_{\text{normal}} = \frac{1}{|P|} \sum_{p \in P} \left(1 - \left\langle \frac{\nabla f_{\text{HDR}}(p)}{\|\nabla f_{\text{HDR}}(p)\|}, \mathbf{n}_{g^*} \right\rangle \right)^2. \quad (11)$$

Moreover, to ensure that each Gaussian approximates a locally planar surface element, we introduce a disk regularization term that promotes disk-like anisotropic configurations. Rather than directly minimizing the smallest principal axis—which may result in unstable gradients—we adopt a differentiable softmin-based formulation to enable smooth and effective optimization:

$$\mathcal{L}_{\text{disk}} = \frac{1}{|\mathcal{G}|} \sum_{g \in \mathcal{G}} -\tau \cdot \log \left(\sum_{i=1}^3 e^{-s_i^g/\tau} \right), \quad (12)$$

where s_i^g denotes the scale of Gaussian g along its i -th principal axis, and τ is a temperature parameter that controls the sharpness of the softmin approximation. This formulation encourages each Gaussian to adopt a geometrically coherent, anisotropic structure that aligns with the local surface geometry, thereby improving training stability and spatial regularity.

2.3. Joint Training for Harmonizing Optimization

In this stage, the irradiance and geometry fields are jointly refined via a coupled optimization strategy, which is essential for high-fidelity HDR-NVS. Accurate geometric structures guide light propagation near object boundaries and shadows, while radiometric cues help recover fine-grained geometry. Central to this process is a geometry-aware density control mechanism that dynamically adjusts the spatial distribution of Gaussians based on feedback from the exposure-aware

Table 1. Quantitative comparison on synthetic datasets. LDR-OE and LDR-NE represent LDR-NVS settings using exposure subsets $\{t_1, t_3, t_5\}$ and $\{t_2, t_4\}$, respectively. HDR represents NVS based on complete HDR reconstruction. Our method achieves the best performance across all configurations, maintaining robust results under the challenging setting and demonstrating strong generalization.

Method	LDR-OE (t_1, t_3, t_5)			LDR-NE (t_2, t_4)			HDR		
	PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$	PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$	PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$
NeRF (Mildenhall et al., 2021)	15.63	0.612	0.309	17.83	0.652	0.353	—	—	—
3D-GS (Kerbl et al., 2023)	22.37	0.690	0.276	18.97	0.778	0.309	—	—	—
NeRF-W (Martin-Brualla et al., 2021)	29.83	0.936	0.047	29.22	0.927	0.050	—	—	—
HDR-NeRF (Huang et al., 2022)	39.35	0.979	0.028	36.58	0.954	0.028	36.12	0.908	0.019
HDR-GS (Cai et al., 2024)	41.06	0.980	0.013	36.36	0.973	0.019	37.98	0.976	0.014
GaussHDR (Liu et al., 2025)	39.74	0.974	0.021	36.89	0.977	0.018	37.66	0.967	0.025
Mono-HDR (Zhang et al., 2025b)	30.16	0.923	0.051	31.44	0.938	0.055	38.81	0.975	0.014
Expo-GS (Ours)	41.38	0.989	0.010	37.47	0.984	0.014	39.06	0.981	0.010



Figure 4. Qualitative comparisons on synthetic datasets. Our method achieves superior color fidelity and geometric accuracy over baseline 3D-GS, with zoom-in regions revealing detailed improvements, particularly in high-frequency textures and shadow boundaries.

signed distance field f_{HDR} . The growth activation for a candidate Gaussian center $\mathbf{c}$ is defined as:

$$\epsilon_g = \nabla_g + \omega_s \cdot \exp\left(-\frac{f_{\text{HDR}}(\mathbf{c})^2}{2\sigma^2}\right) + \omega_n \cdot (1 - \|\nabla f_{\text{HDR}}(\mathbf{c})\|), \quad (13)$$

where ∇_g is the accumulated training gradient, and the remaining terms encourage growth near the zero-level surface while suppressing updates in uncertain regions. Gaussians with $\epsilon_g > \tau_g$ are duplicated with perturbations to refine confident structures. Conversely, pruning is guided by a reliability-based score:

$$\epsilon_p = \sigma_a - \omega_p \cdot \left(1 - \exp\left(-\frac{f_{\text{HDR}}(\mathbf{c})^2}{2\sigma^2}\right)\right), \quad (14)$$

where σ_a is the accumulated opacity. Gaussians with $\epsilon_p < \tau_p$ are removed from further optimization, reducing

clutter and suppressing artifacts. By integrating growth and pruning, the geometry-aware density control aligns the Gaussian layout with the evolving Expo-SDF, reinforcing consistency between appearance and structure. To enable stable joint training under such dynamic updates, we combine radiance and geometry losses into a unified objective:

$$\mathcal{L}_{\text{joint}} = \mathcal{L}_{\text{render}} + \lambda_{\text{SDF}} \cdot \mathcal{L}_{\text{SDF}}^{\text{HDR}} + \lambda_{\text{normal}} \cdot \mathcal{L}_{\text{normal}} + \lambda_{\text{disk}} \cdot \mathcal{L}_{\text{disk}}, \quad (15)$$

where $\mathcal{L}_{\text{render}}$ supervises radiance reconstruction, and the remaining terms ensure geometric consistency under varying exposures. The Interactive Joint Training strategy enables geometric structures to inform radiance field modeling, while radiometric cues, in turn, refine fine-grained geometric details. This bidirectional guidance significantly enhances both visual fidelity and structural consistency.



Figure 5. Qualitative comparisons on real-world datasets. Our method achieves superior color fidelity and geometric accuracy over others, with zoom-in regions revealing detailed improvements in overexposed highlights and fine textures under challenging illumination.

Table 2. Quantitative comparisons on the LDR-OE real-world.

Method	PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$
NeRF	16.21	0.682	0.295
3D-GS	18.32	0.849	0.113
NeRF-W	29.39	0.914	0.087
HDR-NeRF	32.35	0.939	0.065
HDR-GS	34.94	0.962	0.031
Expo-GS	35.59	0.981	0.020

Table 3. Quantitative comparisons on the LDR-NE real-world.

Method	PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$
NeRF	15.84	0.716	0.227
3D-GS	20.38	0.719	0.161
NeRF-W	29.17	0.918	0.086
HDR-NeRF	32.71	0.947	0.071
HDR-GS	31.24	0.953	0.045
Expo-GS	32.17	0.972	0.033

3. Experiments

3.1. Experimental Settings

Dataset. We conduct our experiments based on HDR-NeRF (Huang et al., 2022), a dataset designed for NVS under HDR conditions. This dataset features multi-view, multi-exposure captures across 8 synthetic scenes and 4 real-world scenes. Each scene is recorded from 35 distinct viewpoints, with five different exposure levels per view. Each synthetic scene is accompanied by a corresponding HDR reference stored in the OpenEXR (.exr) format. Following prior work (Cai et al., 2024; Wu et al., 2024c), we use HDR images from 18 viewpoints for training, each paired with one randomly sampled LDR exposure from $\{t_1, t_3, t_5\}$. The remaining 17 viewpoints are reserved for evaluation.

Experimental Details. Our framework is trained in three stages. The initialization of Gaussian point parameters follows the settings in 3D-GS (Kerbl et al., 2023), while the tone mapping network is adopted from HDR-GS (Cai et al., 2024). We first perform 8,000 iterations of 3D-GS-based warm-up to establish an initial discrete radiance field. This is followed by 12,000 iterations of implicit training on the Expo-SDF to refine the geometry and extract a high-fidelity surface point cloud. Finally, we conduct 10,000 iterations of

joint optimization to balance radiance accuracy and geometric consistency. The entire training pipeline is implemented in PyTorch and optimized with Adam (Zhang et al., 2024b), running for a total of 30,000 iterations on a single NVIDIA RTX A6000 GPU. See Appendix 6 for hyperparameters.

3.2. Results

Quantitative Comparisons on Synthetic Datasets. Table 1 provides a comprehensive quantitative assessment of NVS performance across synthetic datasets under three distinct exposure configurations: LDR with exposure ($\{t_1, t_3, t_5\}$), LDR with exposure ($\{t_2, t_4\}$), and HDR. Our proposed methodology, Expo-GS, demonstrates superior performance compared to existing approaches across virtually all evaluation metrics with the sole exception being PSNR under the LDR-NE configuration. Furthermore, within the HDR domain, Expo-GS exhibits exceptional efficacy, registering a PSNR of 39.06 dB, which exceeds HDR-NeRF (Huang et al., 2022) by 2.94 dB and HDR-GS (Cai et al., 2024) by 1.08 dB. Our method achieves superior perceptual fidelity, evidenced by the lowest LPIPS and highest SSIM, demonstrating its ability to preserve visual authenticity and geometric consistency, validating the effectiveness of our exposure-aware design and joint optimization strategy.

Table 4. Ablation study on training stages highlighting the importance of geometry field training.

Method	LDR-OE (t_1, t_3, t_5)			LDR-NE (t_2, t_4)			HDR		
	PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$	PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$	PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$
Baseline	22.37	0.690	0.276	18.97	0.778	0.309	—	—	—
Irradiance (8K)	29.17	0.912	0.055	29.86	0.914	0.063	28.86	0.894	0.091
Geometry (12K)	31.92	0.929	0.049	31.78	0.923	0.057	30.29	0.915	0.086
Geometry + 1K	36.41	0.961	0.031	35.39	0.955	0.026	34.76	0.963	0.037
Joint (10K)	41.38	0.989	0.010	37.47	0.984	0.014	39.06	0.981	0.010

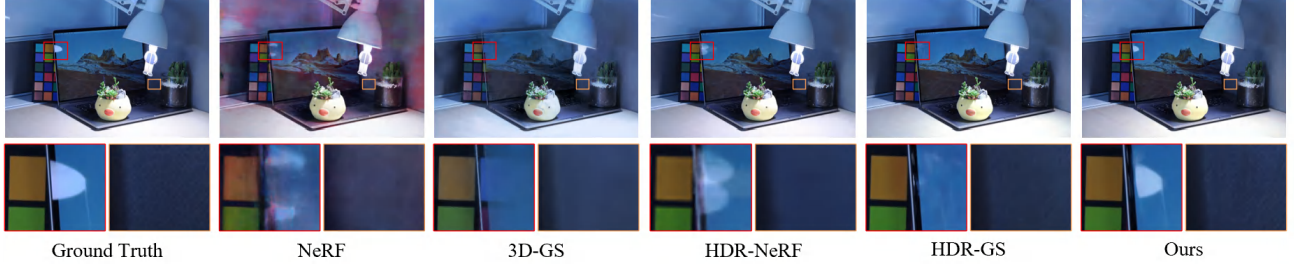


Figure 6. Geometric comparison of different methods in real-world scenes with specular reflections.

Qualitative Comparisons on Synthetic Datasets. Figure 4 presents qualitative comparisons across multiple approaches on synthetic scenes. Traditional NeRF (Mildenhall et al., 2021) and 3D-GS (Kerbl et al., 2023) suffer from severe color smearing and geometric distortions, particularly in regions with challenging illumination. While HDR-NeRF (Huang et al., 2022) and HDR-GS (Cai et al., 2024) enhance visual realism to some extent, they still exhibit noticeable color inconsistencies and geometric blurring under extreme lighting transitions. In contrast, our Expo-GS method reconstructs both radiance and geometry with high fidelity. Fine-grained textures—such as wood grain, fabric patterns, and shadow boundaries—are clearly preserved and visually aligned with the ground truth across all settings.

Quantitative Comparisons on Real-World Datasets. As shown in Table 2 and 3, our Expo-GS framework achieves consistently superior performance across all exposure conditions. Under the LDR-OE setting, it attains a PSNR of 35.59 dB, SSIM of 0.981, and LPIPS of 0.020, significantly outperforming HDR-GS (Cai et al., 2024) and HDR-NeRF (Huang et al., 2022). Even in the more challenging LDR-NE scenario, it maintains robust performance with the highest PSNR of 32.17 dB and the lowest LPIPS of 0.033. The results demonstrate robust generalization and resilience.

Qualitative Comparisons on Real-World Datasets. Figure 5 further demonstrates qualitative improvements over 3D-GS (Kerbl et al., 2023). In overexposed regions (e.g., light bulbs, duck), our method suppresses saturation and geometric distortion in overexposed regions and recovers fine detail in underexposed areas; exposure-guided supervision further enhances high-frequency reconstruction, collectively

reducing radiometric bias and structural artifacts under challenging real-world settings, showing strong applicability.

Ablation study. Table 5 shows that nonlinear CRFs consistently outperform the linear variant, with the logarithmic transform achieving the best PSNR (41.33 and 37.18 under LDR), highlighting its effectiveness in contrast enhancement. We further analyze exposure-time combinations and observe that increasing exposure diversity consistently improves performance. As shown in Table 4, once a pseudo-SDF is established, early iterations of the third-stage training (Geometry+1k) yield immediate performance gains, indicating that SDF quality directly affects both photometric fidelity and geometric consistency, and confirming the critical intermediary role of the second stage in our pipeline.

Table 5. Ablation study on distinct CRF and exposure time sets.

CRF Domain	Linear	Logarithmic	Reinhard	ACES
HDR	29.57	38.77	38.91	38.82
LDR-OE	33.49	41.33	40.14	40.65
LDR-NE	32.82	37.18	36.82	36.64

Exposure	$\{t_3\}$	$\{t_1, t_5\}$	$\{t_1, t_3, t_5\}$	$\{t_1, \dots, t_5\}$
HDR	24.17	32.81	38.64	38.98
LDR-OE	23.78	35.44	41.66	41.69
LDR-NE	23.13	34.25	36.81	37.53

Geometric comparisons. Figure 6 demonstrates a challenging scene with strong specularities, where Expo-GS accurately reconstructs both mirror-like and diffuse surfaces while preserving fine details and non-Lambertian effects. Figure 7 highlights the impact of geometry training, showing improved structural accuracy and contour clarity via

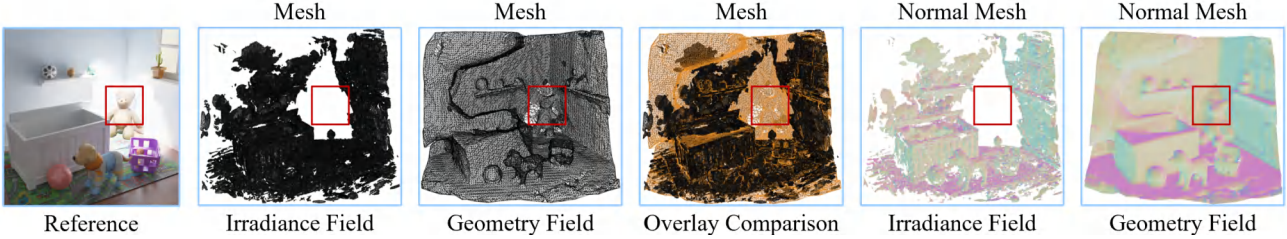


Figure 7. Geometric comparisons of training after first and second stage in synthetic scenarios.

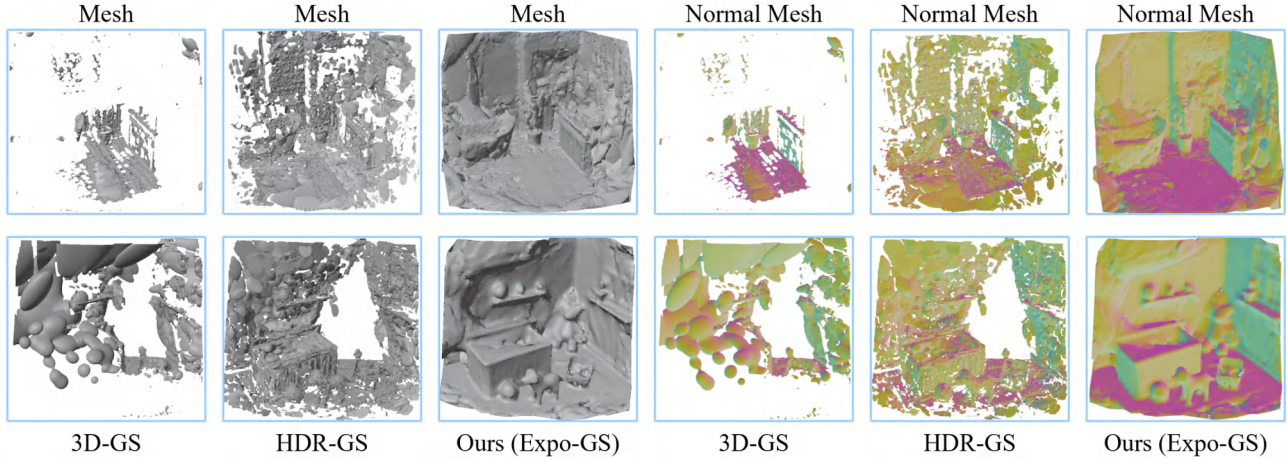


Figure 8. Geometric comparisons of different methods in synthetic scenarios using mesh visualization.

refined point distributions. Compared with prior methods in Figure 8, 3D-GS yields noisy, coarse geometry, and HDR-GS suffers from geometric instability, whereas our method achieves smoother surfaces, more accurate geometry, and better normal consistency. More results are in Appendix F.

4. Related work

3D Gaussian Splatting (3D-GS). 3D-GS has attracted wide attention for real-time (Qu et al., 2024; Wang et al., 2024; Hyun & Heo, 2024; Wu et al., 2024b; Peng et al., 2024) and differentiable rendering (Feng et al., 2025; Lee et al., 2024; Yao et al., 2024), with rapid progress driven by hierarchical pruning and cross-domain extensions (Yu et al., 2024; Chen & Lee, 2024; Zhu et al., 2024). However, existing 3D-GS methods struggle in HDR-NVS (Kerbl et al., 2023; Cai et al., 2024). Their reliance on spherical harmonics limits modeling of extreme luminance variations (Jiang et al., 2024; Liang et al., 2024), and the lack of exposure-aware geometric supervision often leads to structural artifacts and loss of fine details (Wu et al., 2024c; Zhang et al., 2025a).

Signed Distance Function (SDF). Neural networks can represent 3D geometry using continuous SDFs (Chou et al., 2023; Park et al., 2019; Sitzmann et al., 2020; Choi et al., 2024), where surfaces are defined as the zero-level set of an implicit field (Zhang et al., 2024a; Yu et al., 2024; Choi

et al., 2024). Combining SDFs with volumetric rendering (e.g., VolSDF (Yariv et al., 2021), NeuS (Wang et al., 2021)) or 3D Gaussian splatting methods (e.g., SuGaR (Guédon & Lepetit, 2024), GSDF (Yu et al., 2024)) improves surface sharpness and structural coherence (Zakharov et al., 2020; Mu et al., 2021; Chen et al., 2024; Cao & Taketomi, 2024). However, most existing SDF-based methods assume fixed-exposure LDR inputs and are exposure-agnostic, leading to inaccurate geometry in over- or under-exposed regions and visible artifacts in NVS. This highlights the lack of exposure-aware SDF modeling for HDR-NVS. For additional related work, refer to Appendix C for additional discussion.

5. Conclusion

We present Expo-GS, a biologically inspired HDR-NVS framework that jointly models radiance and geometry across varying exposures. By combining Expo-SDF with a decoupled-then-joint training strategy, our method mitigates radiometric bias and geometric inconsistency under extreme illumination. Experiments on both synthetic and real-world datasets demonstrate that Expo-GS achieves state-of-the-art performance while maintaining superior perceptual quality and geometric consistency. To our knowledge, this is the first HDR-NVS framework that unifies multi-exposure radiometric and geometric modeling, offering new insights into exposure-aware scene reconstruction.

Limitations

Although Expo-GS performs well on HDR-NVS, it also has limitations. Our joint optimization introduces additional training overhead because of the multi-stage schedule and the Expo-SDF refinement. The moderate increase in training time mainly comes from the second stage (Geometry Field Training). However, when combined with the single-pass rendering strategy in the first stage and the lightweight optimization in the third stage, the overall training time remains reasonable, and our method still outperforms other methods in terms of inference speed. As shown by the main HDR-NVS experiments, our method achieves a favorable balance between performance and efficiency, where “(8 / 21 / 7)” denotes the relative training cost of the three stages. Additionally, the reliance on multi-exposure inputs requires strictly static scenes, which limits applicability to dynamic settings. Future work will explore extensions to time-varying scenes in real-world environments.

Impact Statement

This paper presents Expo-GS, a novel framework for high dynamic range novel view synthesis that advances the field of 3D scene reconstruction and rendering. Our work has several potential societal implications worth considering.

Positive Impacts. The proposed method enables more faithful capture and reconstruction of real-world scenes under challenging lighting conditions, which could benefit applications in virtual reality, autonomous systems, digital content creation, and cultural heritage preservation. By improving geometric consistency and radiometric fidelity in HDR environments, Expo-GS may enhance the quality of immersive experiences and enable more accurate 3D documentation of physical spaces. The biologically-inspired design philosophy, drawing from human visual system principles, also contributes to the broader understanding of perception-aware computational imaging.

Environmental Considerations. Our method achieves competitive inference speed (131 fps) while maintaining reasonable training costs on a single GPU, representing a relatively efficient approach compared to some alternatives. However, we acknowledge that continued research in neural rendering contributes to overall computational resource consumption, and we encourage future work to further optimize efficiency.

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A. Appendix

B. Theoretical Derivations

B.1. Analysis of Equation (8)

Equation (8) defines the *exposure-normalized Gaussian density*:

$$\dot{d}(\mathbf{p}) = \sum_j \frac{\alpha_j}{E(\boldsymbol{\mu}_j) + \varepsilon} \exp\left[-\frac{1}{2}(\mathbf{p} - \boldsymbol{\mu}_j)^\top \boldsymbol{\Sigma}_j^{-1}(\mathbf{p} - \boldsymbol{\mu}_j)\right], \quad (8)$$

where each Gaussian's opacity coefficient α_j is scaled by the estimated exposure $E(\boldsymbol{\mu}_j)$ at its center, with a small constant ε added to ensure numerical stability. This exposure-aware attenuation preserves the influence of well-exposed regions while down-weighting overexposed highlights and severely underexposed, noise-prone areas.

The resulting density field $\dot{d}(\mathbf{p})$ is strictly positive, infinitely differentiable, and globally bounded. Moreover, it is monotonically *decreasing* with increasing exposure, enabling robust structural modeling in HDR conditions. Notably, the scaled negative logarithm of $\dot{d}(\mathbf{p})$ induces a quadratic form in $\mathbf{p}$, which serves as the foundation for the pseudo-signed distance function defined in Equation (9) and utilized in geometric supervision.

B.2. Proof of Equation (8)

1. Positivity and smoothness.

Each term in the summation is positive: $\alpha_j > 0$, $E(\boldsymbol{\mu}_j) + \varepsilon > 0$, and the Gaussian exponential kernel is positive and C^∞ . Therefore, $\dot{d}(\mathbf{p})$ is strictly positive and infinitely differentiable over $\mathbb{R}^3$.

2. Global upper bound.

Since the denominator is lower-bounded by ε , we have:

$$\dot{d}(\mathbf{p}) \leq \sum_j \frac{\alpha_j}{\varepsilon},$$

ensuring $\dot{d}(\mathbf{p})$ is globally bounded and $\log \dot{d}(\mathbf{p})$ is finite.

3. Monotonicity with respect to exposure.

Differentiating each term w.r.t. $E(\boldsymbol{\mu}_j)$:

$$\frac{\partial}{\partial E(\boldsymbol{\mu}_j)} \left(\frac{\alpha_j}{E(\boldsymbol{\mu}_j) + \varepsilon} e^{-\frac{1}{2}(\dots)} \right) = -\frac{\alpha_j}{(E(\boldsymbol{\mu}_j) + \varepsilon)^2} e^{-\frac{1}{2}(\dots)} < 0,$$

indicating $\dot{d}(\mathbf{p})$ decreases monotonically with local exposure.

4. Local dominance.

When a single Gaussian g_k dominates locally:

$$\dot{d}(\mathbf{p}) \approx \frac{\alpha_k}{E_k + \varepsilon} \exp\left[-\frac{1}{2}(\mathbf{p} - \boldsymbol{\mu}_k)^\top \boldsymbol{\Sigma}_k^{-1}(\mathbf{p} - \boldsymbol{\mu}_k)\right], \quad E_k := E(\boldsymbol{\mu}_k).$$

5. Quadratic structure of the log-density.

Taking the negative logarithm:

$$-2 \log \dot{d}(\mathbf{p}) = (\mathbf{p} - \boldsymbol{\mu}_k)^\top \boldsymbol{\Sigma}_k^{-1}(\mathbf{p} - \boldsymbol{\mu}_k) + C,$$

where $C = -2 \log \alpha_k + 2 \log(E_k + \varepsilon)$ is constant in $\mathbf{p}$.

6. Euclidean upper bound.

Let $\boldsymbol{\Sigma}_k = \text{diag}(s_{k1}^2, s_{k2}^2, s_{k3}^2)$, and define $s_{\min} := \min_i s_{ki}$. Then:

$$-2 \log \dot{d}(\mathbf{p}) \geq \frac{\|\mathbf{p} - \boldsymbol{\mu}_k\|_2^2}{s_{\min}^2}.$$

7. Gradient alignment.

The gradient is:

$$\nabla_{\mathbf{p}} \dot{d} = \dot{d} \Sigma_k^{-1} (\boldsymbol{\mu}_k - \mathbf{p}),$$

which aligns with the Gaussian's outward normal, enabling normal-aligned supervision.

8. Conclusion.

Equation (8) defines a well-behaved, exposure-aware surrogate density. Its negative logarithm induces a Mahalanobis distance, forming the mathematical basis for the pseudo-SDF used in Equation (9).

B.3. Analysis of Equation (9)

Equation (9) defines a *pseudo-signed distance function (pseudo-SDF)*:

$$f_{\text{HDR}}(\mathbf{p}) = \pm s_{g^*} \sqrt{-2 \log \dot{d}(\mathbf{p})}, \quad (9)$$

where g^* denotes a locally dominant Gaussian near p , and s_{g^*} is a local scale parameter derived from its principal axes (e.g., lower-bounded by its smallest axis length). This formulation transforms the log-density into a distance-like scalar field whose sign is determined by the orientation of the local surface normal. Importantly, g^* and s_{g^*} are introduced here as analytical devices for interpreting the smooth potential $-\log \dot{d}(p)$; in our implementation, $f_{\text{HDR}}(p)$ is evaluated directly from $\dot{d}(p)$ without differentiating through any hard nearest-Gaussian or min-axis selection. Consequently, $f_{\text{HDR}}(\mathbf{p})$ behaves like a true SDF in well-exposed regions and provides reliable geometric cues in radiometrically ambiguous areas.

B.4. Proof of Equation (9)

1. Dominant Gaussian assumption.

Assume g^* dominates in the local neighborhood:

$$\dot{d}(\mathbf{p}) \approx \frac{\alpha_{g^*}}{E(\boldsymbol{\mu}_{g^*}) + \varepsilon} \exp\left[-\frac{1}{2}(\mathbf{p} - \boldsymbol{\mu}_{g^*})^\top \Sigma_{g^*}^{-1}(\mathbf{p} - \boldsymbol{\mu}_{g^*})\right].$$

2. Log transformation.

Taking $-2 \log$:

$$-2 \log \dot{d}(\mathbf{p}) = (\mathbf{p} - \boldsymbol{\mu}_{g^*})^\top \Sigma_{g^*}^{-1}(\mathbf{p} - \boldsymbol{\mu}_{g^*}) + C.$$

3. Principal axis decomposition.

Diagonalizing Σ_{g^*} :

$$(\mathbf{p} - \boldsymbol{\mu}_{g^*})^\top \Sigma_{g^*}^{-1}(\mathbf{p} - \boldsymbol{\mu}_{g^*}) = \sum_{i=1}^3 \frac{\Delta_i^2}{s_{g^*,i}^2}.$$

4. Distance upper bound.

Let $s_{g^*} = \min_i s_{g^*,i}$. Then:

$$-2 \log \dot{d}(\mathbf{p}) \geq \frac{\|\mathbf{p} - \boldsymbol{\mu}_{g^*}\|_2^2}{s_{g^*}^2}.$$

5. Distance approximation.

Rescaling gives:

$$s_{g^*} \sqrt{-2 \log \dot{d}(\mathbf{p})} \approx \|\mathbf{p} - \boldsymbol{\mu}_{g^*}\|_2.$$

6. Sign function.

Using the local surface normal $\mathbf{n}_{g^*}$:

$$\text{sign}(f_{\text{HDR}}(\mathbf{p})) = \text{sign}((\mathbf{p} - \boldsymbol{\mu}_{g^*}) \cdot \mathbf{n}_{g^*}).$$

7. Definition of pseudo-SDF.

Combining the above, we define:

$$f_{\text{HDR}}(\mathbf{p}) = \pm s_{g^*} \sqrt{-2 \log \dot{d}(\mathbf{p})}.$$

8. Gradient alignment.

Given

$$\nabla_{\mathbf{p}} \dot{d} = \dot{d} \Sigma_{g^*}^{-1}(\boldsymbol{\mu}_{g^*} - \mathbf{p}),$$

the gradient ∇f_{HDR} is aligned with the local normal $\mathbf{n}_{g^*}$.

9. Conclusion.

The function $f_{\text{HDR}}(\mathbf{p})$ behaves as a smooth, exposure-aware approximation of a signed distance field, suitable for geometry regularization under HDR settings.

Whether a region is up-weighting or down-weighting therefore depends jointly on α_j and $\mathcal{E}(\mu_j)$, rather than on $\mathcal{E}(\mu_j)$ alone. In overexposed regions, $\mathcal{E}(\mu_j)$ becomes large, so it appears in the denominator and strongly attenuates the contribution of the corresponding Gaussians. In underexposed regions, however, α_j becomes very small: in regions with extreme underexposure, the low-signal conditions lead to unstable gradients, and the learned Gaussian opacities tend to be close to zero. As a result, the ratio $\frac{\alpha_j}{\mathcal{E}(\mu_j) + \epsilon}$ becomes very small (with $\epsilon > 0$ fixed), so the contribution of underexposed Gaussians is also effectively suppressed.

C. Extended Related Work

High Dynamic Range (HDR) Imaging. Traditional HDR imaging hinges on multi-exposure fusion to reconstruct the full luminance spectrum of a scene. Early methods focused on radiometric calibration to counteract nonlinear camera responses (Zhang et al., 2025b; Liu et al., 2025; Huang et al., 2024; Jun-Seong et al., 2022; Wu et al., 2024a), while recent works integrate deep learning with physical sensor models, using attention-based fusion and physics-inspired tone mapper for enhanced fidelity (Seetzen et al., 2023; Zhang & Yau, 2009). HDR-NeRF (Huang et al., 2022) introduces HDR capabilities into Neural Radiance Fields via inverse gamma correction and HDR-aware training, improving reconstructions in low-light regions. HDR-GS (Cai et al., 2024) further advances this by adapting 3D-GS (Kerbl et al., 2023) to HDR synthesis, leveraging dual dynamic range point cloud modeling and differentiable rasterization for simultaneous HDR-LDR reconstruction with exposure control. However, most HDR methods emphasize pixel-wise radiometric accuracy and rely heavily on color regression, overlooking the intricate coupling between lighting variations and geometry—especially problematic in 3D scenes where structural consistency is crucial.

Human Visual System (HVS). HVS possesses a remarkable ability to perceive both color and structural information of scenes under highly dynamic and varying lighting conditions (Thorpe et al., 1996). This capacity arises from several key perceptual mechanisms (Parraga et al., 2000; Banks et al., 2012).

First, exposure adaptation allows the human eye to function across a wide luminance range—from dim starlight at night to bright sunlight at noon—through dynamic gain control by photoreceptor cells in the retina (Goodale & Haffenden, 1998). Second, contrast sensitivity drives the visual system to focus on local edges and changes in luminance rather than absolute pixel intensity (Adini et al., 2002). This mechanism is essential for capturing object boundaries and geometric structure. Third, the HVS performs dual-stream processing to separately process color and structure (Field et al., 1993). Specifically, cone cells in the retina transmit fine-grained chromatic and texture information through the parvocellular pathway (color stream) to the visual cortex, while rod cells are more sensitive to motion and coarse structure, channeling information through the magnocellular pathway (structure stream) (Goebel et al., 2004; Salin & Bullier, 1995). These two streams are later recombined in higher-level visual areas to construct a coherent and stable perception of the visual world (Toosy et al., 2004; Lu & Sperling, 2001).

Inspired by the HVS, a more robust approach is to decouple color (i.e., irradiance) and structure (i.e., geometry) during the training process. Specifically, radiometric and geometric attributes of the scene are modeled separately and then jointly optimized at the perceptual level. This mimics the human strategy of treating different exposure regions with varying sensitivity and integrating information based on local reliability. Thus, the perceptual principles of the HVS provide a biologically grounded motivation for decoupled modeling in HDR-NVS. Emulating the HVS paradigm of “first decomposing, then integrating” enables the development of HDR-NVS systems that are not only photometrically faithful but also geometrically stable across varying lighting conditions.

D. In-Depth Design Analysis

In multi-exposure or extreme dynamic range scenarios, geometry optimized solely from pixel color residuals often fails. First, gradients from saturated or underexposed pixels are either zero or unstable, causing the density σ to expand indiscriminately and leading to surface drift. Second, exposure times (Δt) vary across views; after the camera’s nonlinear response function, residual magnitudes become imbalanced, further compounding depth inconsistencies.

These issues are well-documented in prior work: *RawNeRF* demonstrates that NeRF fails to recover highlights and shadows under LDR conditions, while *Gaussian-DK* reports severe ghosting artifacts in *3D-GS* under high-contrast lighting. In summary, ignoring exposure reliability and relying solely on color cues hinders robust geometric reconstruction under varying exposure conditions—an observation supported by both physical modeling and empirical evidence.

Although the exposure time Δt is a global setting, the local exposure conditions within a single frame are far from uniform; bright highlights and deep shadows often coexist in our dataset. As early as the HDR fusion method of Debevec&Malik (1997), the pixel weight $w(z)$ was given a bell-shaped profile to down-weight overexposed and underexposed samples. The spirit is identical here: we assess pixel-wise confidence from exposure, rather than just blindly using the global Δt .

In the bathroom scene of the synthetic dataset, the red flowerpot provides a representative test case. Under normal exposure (t_3), the average RGB values are approximately $P_a \approx (161, 74, 58)$, with the red channel at 161. Upon inspection, we observed that these pixels retain full weight—and in some cases even receive slightly increased weight—during subsequent computation and training. Under overexposure (t_5), the same region exhibits average RGB values of $P_b \approx (222, 170, 144)$, with the red channel rising to 222. In this case, the SDF weight assigned to P_b is slightly reduced, which is appropriate given that local highlights introduce noisy color residuals that can destabilize pseudo-SDF supervision.

If a large, uniform area has a high $\max(\text{RGB})$ value, it is beneficial to reduce the weight of its contribution to the SDF. Consider an area that is almost pure red (255, 0, 0) and contains little geometric or textural detail. Assigning the sparsely distributed 3D point cloud to this area is sufficient. Reducing its density does not affect the appearance, and the boundary geometry is still defined by the surrounding well-exposed pixels. Therefore, the weighting scheme does not misclassify vibrant colors; on the contrary, it highlights their geometric value and contributes to more stable optimization. And our goal is not to calculate the precise absolute brightness of pixels but to quickly determine whether a pixel is within a reliable (neither overexposed/underexposed) range for subsequent SDF geometric gradient weighting.

E. Hyperparameter Configuration for Expo-GS Training

We detail the hyperparameter configuration used across the three stages of Expo-GS: irradiance field training, geometry field training, and interactive joint training.

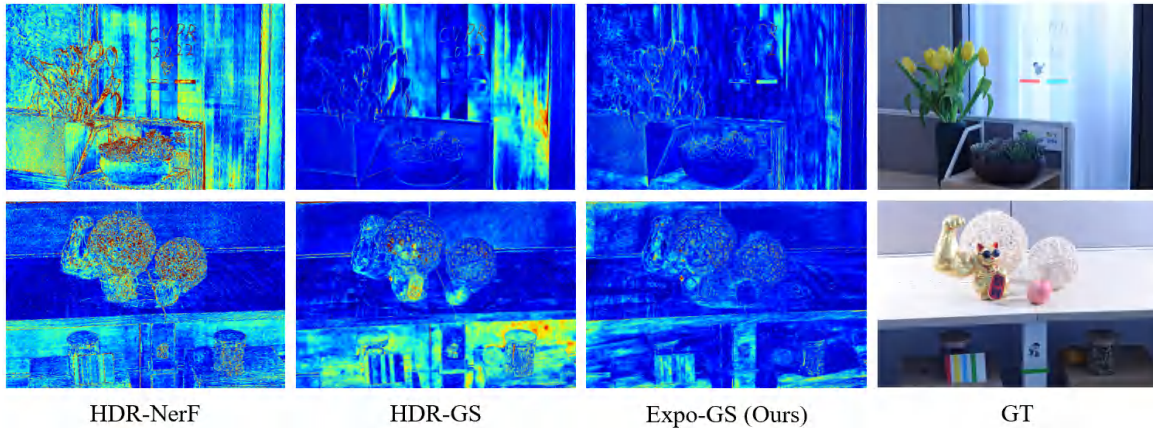


Figure 9. Comparison of error maps with state-of-the-art methods.

Parameter	Value	Description
1. General Training Settings		
Optimizer	Adam	Used in all training stages
Learning Rate (Stage 1)	2.5e-3	Irradiance field training
Learning Rate (Stage 2)	1.0e-3	Geometry field training
Learning Rate (Stage 3)	5.0e-4	Joint optimization
Total Iterations	30,000	8000 (Stage 1) + 12000 (Stage 2) + 10000 (Stage 3)
Hardware	NVIDIA RTX A6000	49 GB VRAM (49140 MiB), Single-GPU
Framework	PyTorch (Imambi et al., 2021)	CUDA 11.8 compatible (Paszke et al., 2019)
2. Geometry Field Training Settings		
Expo-SDF Loss	$\lambda_{\text{SDF}} = 0.2$	Geometry field stage
Normal Alignment Loss	$\lambda_{\text{normal}} = 0.2$	Geometry field stage
Disk Regularization Loss	$\lambda_{\text{disk}} = 0.1$	Geometry field stage
3. Joint Optimization Settings		
Expo-SDF Loss	$\lambda_{\text{SDF}} = 0.1$	Joint stage
Normal Alignment Loss	$\lambda_{\text{normal}} = 0.1$	Joint stage
Disk Regularization Loss	$\lambda_{\text{disk}} = 0.05$	Joint stage

Table 6. Hyperparameter configuration across training stages for Expo-GS, comprising Stage 1 (irradiance field training), Stage 2 (geometry field training), and Stage 3 (interactive joint training).

F. Extended Results

Tone mapper. The core limitation of fixed tone mappers lies in their generalizability. Their parameters are empirical, and when faced with cross-scene or extreme exposure conditions, manual re-tuning is often required to avoid color bias and detail loss. In contrast, learnable tone mappers can adaptively map relationships end-to-end without human intervention. Furthermore, nearly all previous HDR-NVS methods (HDR-NeRF, HDR-GS, etc.) inherently rely on learnable tone mappers. If we were to manually select the optimal fixed tone mappers for each of them, it would introduce new unfair factors. Therefore, we retain the learnable tone mapper to maintain end-to-end properties and ensure that comparisons with existing methods focus on exposure modeling capabilities rather than the tone mapper itself.

Evaluation. To evaluate the performance of HDR-NVS, we adopt three widely used metrics: Peak Signal-to-Noise Ratio (PSNR) (Johnson, 2006), Structural Similarity Index Measure (SSIM) (Wang, 2004), and Learned Perceptual Image Patch Similarity (LPIPS) (Zhang et al., 2018). PSNR (Johnson, 2006) directly quantifies radiometric fidelity through pixel-level differences, while SSIM (Wang, 2004) and LPIPS (Zhang et al., 2018) indirectly reflect geometric coherence by measuring structural and perceptual similarity, respectively. Together, these metrics provide a dual-perspective assessment that captures both irradiance accuracy and geometric consistency.

Method	LDR-OE (t_1, t_3, t_5)			LDR-NE (t_2, t_4)			HDR		
	PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$	PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$	PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$
LTM-NeRF (Huang et al., 2024)	39.42	0.976	0.033	36.67	0.964	0.027	36.55	0.953	0.018
HDR-Plenoxels (Jun-Seong et al., 2022)	36.38	0.959	0.045	35.02	0.943	0.045	33.86	0.941	0.046
Expo-GS (Ours)	41.38	0.989	0.010	37.47	0.984	0.014	39.06	0.981	0.010

Table 7. More comparisons with additional methods.

More comparisons with additional methods. All comparison methods are based on the training and testing set partitions declared in Section 4.1 of our paper. Experimental results 7 demonstrate that our method still achieves a significant lead, outperforming additional methods in both overall reconstruction quality and perceptual consistency across LDR-OE, LDR-NE, and HDR settings.

Comparisons with Vanilla SDF. The experiments 8 demonstrated that the vanilla SDF fails to deliver satisfactory results and cannot reasonably handle the exposure conditions of HDR-NVS. We attribute this to the fact that the vanilla SDF forced constraints fail to properly process exposure information, leading to field geometry conflicts. This causes conflicts between irradiance field and geometry field especially in the overexposed and underexposed regions. This experiment

Expo-GS: Exposure-Aware Signed Distance Function in Gaussian Splatting for High Dynamic Range

Method	LDR-OE (t_1, t_3, t_5)			LDR-NE (t_2, t_4)			HDR		
	PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$	PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$	PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$
Expo-GS (w/ Vanilla SDF)	36.59	0.954	0.038	34.36	0.948	0.044	32.95	0.947	0.046
Expo-GS (w/ Vanilla SDF)	36.59	0.954	0.038	34.36	0.948	0.044	32.95	0.947	0.046
Expo-GS (w/ Expo-SDF)	41.38	0.989	0.010	37.47	0.984	0.014	39.06	0.981	0.010

Table 8. Comparisons with Vanilla SDF.

further demonstrates the effectiveness of Expo-SDF.

Method	LDR-OE (t_1, t_3, t_5)			LDR-NE (t_2, t_4)			HDR		
	PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$	PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$	PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$
Standard SDF (Guédon & Lepetit, 2024)	31.52	0.928	0.061	31.27	0.924	0.065	30.71	0.922	0.079
SuGaR (Guédon & Lepetit, 2024)	36.10	0.954	0.049	35.28	0.958	0.049	33.36	0.943	0.048
GSDF (Yu et al., 2024)	38.76	0.966	0.026	36.39	0.971	0.022	34.81	0.957	0.033
PulledGS (Zhang et al., 2024a)	39.69	0.971	0.019	36.07	0.965	0.027	35.95	0.962	0.025
Expo-GS (Ours)	41.38	0.989	0.010	37.47	0.984	0.014	39.06	0.981	0.010

Table 9. Comparisons with SDF-based methods reveal their limitations under LDR inputs.

Comparisons with SDF-based methods. As shown in Table 9, existing SDF-based methods (e.g., SuGaR (Guédon & Lepetit, 2024), GSDF (Yu et al., 2024), and PulledGS (Zhang et al., 2024a)), though effective under fixed-exposure LDR conditions, exhibit suboptimal performance in HDR-NVS scenarios. These methods typically employ dense surface guiding while uniformly weighting multi-exposure inputs, neglecting the exposure disparities inherent in LDR observations. Consequently, they fail to handle the challenging conditions inherent in HDR configurations, leading to the emergence of spurious geometries and degraded radiometric fidelity. Notably, our proposed Expo-GS framework effectively accommodates multi-exposure inputs by incorporating exposure-aware optimization, achieving superior results across all evaluated metrics.

Method	LDR-OE (t_1, t_3, t_5)			LDR-NE (t_2, t_4)			HDR		
	PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$	PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$	PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$
NeRFactor (Zhang et al., 2021)	33.16	0.941	0.038	18.62	0.761	0.325	—	—	—
ReLight-NeRF (Toschi et al., 2023)	36.58	0.957	0.029	21.31	0.819	0.292	—	—	—
GaSLight (Bolduc et al., 2025)	34.65	0.943	0.031	29.89	0.881	0.137	30.44	0.926	0.077
Luminance-GS(Cui et al., 2025)	26.77	0.853	0.071	19.21	0.752	0.304	—	—	—
Expo-GS(Ours)	41.38	0.989	0.010	37.47	0.984	0.014	39.06	0.981	0.010

Table 10. Comparisons with relightable methods reveal their limitations under LDR inputs.

Table 10 reveals that under the LDR-OE setting (exposure levels observed during training), both NeRFactor and ReLight-NeRF deliver reasonable performance owing to their architectural illumination modeling and the availability of ground-truth supervision. The learned mappings between brightness, geometry, and material remain effective within the trained exposure range, yielding relatively high-quality reconstructions. However, under the more challenging LDR-NE setting (unseen exposures), both methods suffer a substantial degradation in performance. This drop is attributed to the absence of explicit exposure modeling, which compels the networks to implicitly approximate brightness differences across views. Consequently, unseen exposure times induce shifts in activation distributions, leading to poor generalization and a sharp decline in image quality.

While NeRFactor and ReLight-NeRF are effective for modeling illumination variations within LDR conditions, their outputs are ultimately governed by apparent brightness. As such, they remain competent for LDR relighting tasks but are unsuitable for HDR-NVS or cross-exposure generalization.

Proxy	LDR-OE $\uparrow$	LDR-NE $\uparrow$	HDR $\uparrow$
$e_i \cdot \text{mean}_c I_{i,c}$	41.12	37.27	38.88
$e_i \cdot (0.299R + 0.587G + 0.114B)$	41.27	37.22	38.96
$e_i \cdot \hat{w}(\max_c I_{i,c})$	40.26	37.03	38.57
$e_i \cdot \max_c I_{i,c}$ (Ours)	41.38	37.47	39.06

Method	NC $\uparrow$	Lap $\downarrow$	NME $\downarrow$
3DGS	0.798	0.033	0.15
HDR-GS	0.852	0.018	0.07
Expo-GS	0.975	0.011	0.01

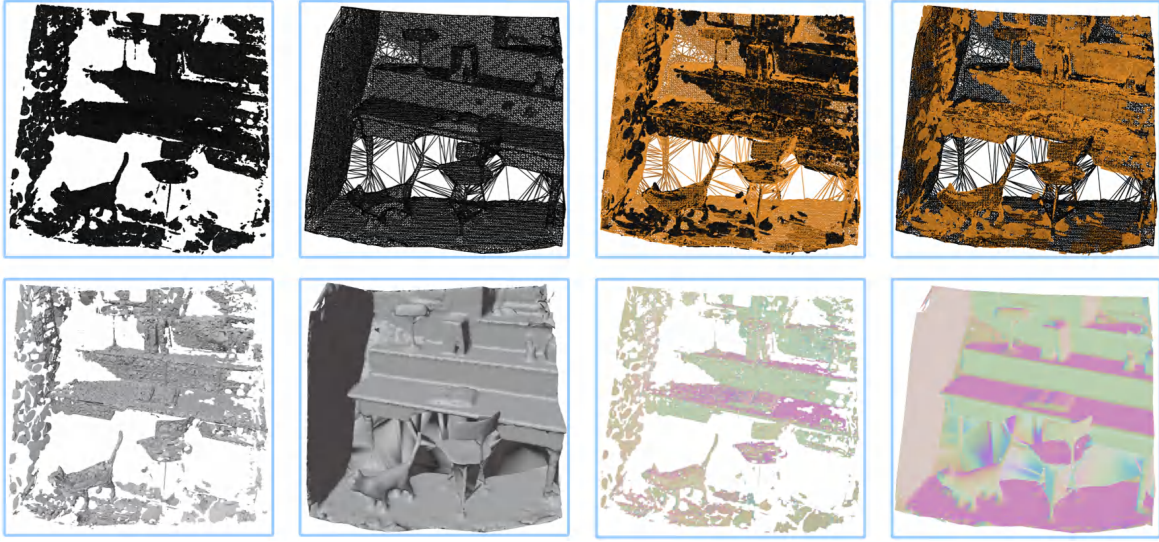


Figure 10. Geometric comparisons of training after first and second stage in synthetic scenarios.

Generalization of exposure distribution. To evaluate the model under more extreme exposure conditions, we first use the learned tone mapper to synthesize LDR images at two additional extreme exposure times, $t_6 = 1/16$ s and $t_7 = 1/32$ s. The framework reconstructs an HDR representation of the scene, scales this HDR signal by the desired exposure factor, and then passes it through the learned tone mapper to obtain an LDR image corresponding to an exposure level never seen during training (see Eq. 3 in the main paper). We then conduct experiments using these synthetically generated LDR images under extreme exposure conditions, and summarize the results on the synthetic dataset in terms of PSNR in Table 11.

When the input comprises only two extreme exposure levels, $\{1/32$ s, 32 s $\}$ (i.e., $\{t_7, t_5\}$), the model still achieves a PSNR of 32–35 dB across all three evaluation settings (HDR, LDR-OE, and LDR-NE), which is generally regarded as visually acceptable. As the exposure interval decreases, the PSNR increases steadily, indicating that reconstruction quality improves as the exposure gap narrows. Nevertheless, even the most extreme two-exposure configuration yields acceptable results (> 30 dB). These experiments demonstrate the robustness of our method across a broad range of exposure conditions and confirm its ability to handle exposure variations even in extreme scenarios.

Validating the significance of the Geometry Field Stage.

From Table 4, we observe that once the pseudo-SDF has been established, the initial iterations of the third stage (Geometry+1k training) yield a noticeable improvement in performance metrics. This suggests that the quality of SDF learning indirectly influences the photometric fidelity and geometric consistency of the synthesized views. As a geometric prior, it plays an essential intermediary role in the overall framework, underscoring the significance of the geometry field training stage.

In fact, the three-stage training is a common GS pipeline. In our experiments, a 12k-iteration geometry field stage serves as a

Metadata Condition	LDR-OE $\uparrow$	LDR-NE $\uparrow$	HDR $\uparrow$
No metadata ($e_i = 1$)	41.05	37.19	38.65
Noisy metadata (+20%)	41.33	37.44	39.04
Full metadata (Ours)	41.38	37.47	39.06

Exposure	t_3	$\{t_7, t_5\}$	$\{t_6, t_5\}$	$\{t_1, t_5\}$	$\{t_1, t_3, t_5\}$	$\{t_1, \dots, t_5\}$
HDR	24.17	32.14	33.58	32.81	38.64	38.98
LDR-OE	23.78	34.93	35.37	35.44	41.66	41.69
LDR-NE	23.13	33.61	33.98	34.25	36.81	37.53

Table 11. Generalization of exposure distribution.

stable configuration for all scenarios, including complex synthetic and real-world environments. We found that convergence is usually achievable without 12k. Table 12 summarizes the results of training our Geometry Field Stage for 3k, 6k, 9k, and 12k iterations.

As shown in our additional ablation on the number of iterations in the Geometry Field stage, increasing the geometry training from 3K to 9K iterations leads to substantial improvements, while the performance saturates between 9K and 12K. This demonstrates that (i) the geometry field stage is indeed crucial for converging to high-quality HDR geometry, and (ii) the method is not overly sensitive to the exact iteration count once a sufficient number of iterations ($\approx 9K$) is used.

Efficiency Comparison. As shown in Table 13, the moderate increase in training time is due to the second stage (Geometry Field Training). However, combined with the single rendering strategy of the first stage and the lightweight optimization of the third stage, the overall training time is not very long, and our method outperforms other methods in terms of inference speed. In fact, based on the main experimental results of HDR-NVS, our method achieves a balance between performance and efficiency.

Qualitative Comparison. Figure 11 illustrates qualitative results on representative synthetic scenes under challenging illumination. The baseline 3D-GS method exhibits prominent artifacts, such as color bleeding, oversaturation, and structure deformation, especially around high-contrast regions like candles, windows, and shadow boundaries. In contrast, our method (Expo-GS) demonstrates significantly improved radiometric accuracy and structural coherence. Reflective surfaces and high-frequency textures (e.g., wallpaper patterns, fabric details, window frames) are faithfully reconstructed, closely matching the ground truth. These improvements highlight the effectiveness of exposure-aware SDF supervision and joint optimization in mitigating hallucinations and ensuring geometric stability across varying exposure conditions.

Geometric comparison. As illustrated in the Figure 10, we provide qualitative visualizations of both the mesh and the corresponding normal maps obtained after the first-stage irradiance field training and the subsequent geometry field training. The irradiance field training alone is insufficient for reliable geometric modeling, as the initial triangular mesh derived from the point cloud displays prominent holes and structural discontinuities. By contrast, the geometry field training, enhanced through Expo-SDF optimization, produces a smoother and more coherent scene geometry. The normal visualizations further highlight the improved geometric fidelity, exhibiting strong surface consistency and continuity. Overall, these qualitative results demonstrate that geometry field training effectively captures structural information and reconstructs scene geometry, mitigating noise and stripe artifacts, filling missing regions, and substantially improving both surface smoothness and normal coherence.

Method	LDR-OE (t_1, t_3, t_5)			LDR-NE (t_2, t_4)			HDR		
	PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$	PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$	PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$
Geometry Field/ 3K	36.83	0.956	0.496	35.66	0.948	0.043	34.37	0.944	0.043
Geometry Field/ 6K	38.32	0.965	0.412	36.18	0.960	0.035	35.88	0.948	0.028
Geometry Field/ 9K	41.41	0.982	0.010	37.43	0.982	0.014	39.01	0.977	0.011
Geometry Field/ 12K	41.38	0.989	0.010	37.47	0.984	0.014	39.06	0.981	0.010

Table 12. Sensitivity analysis of geometry field training.

Method	Training time (min)	GPU memory (GB)	Inference speed (fps)
HDR-NeRF	517	80	0.128
HDR-GS	33	8	122
Expo-GS (ours)	36 (8 / 21 / 7)	6 / 11 / 9	131

Table 13. Training efficiency and inference speed comparison.



Figure 11. Qualitative comparisons on synthetic datasets. Our method achieves superior color fidelity and geometric accuracy over baseline 3D-GS under LDR settings.



Figure 12. Our method achieves superior color fidelity and geometric accuracy under HDR settings.